

READING PASSAGE 3

The Physics of Climate Change Models

A The study of the physics of climate change models has undergone a remarkable transformation over the past several decades. What was once a niche area of academic inquiry has developed into a rich and diverse field of research, drawing on insights from a wide range of disciplines and theoretical traditions.

B The intellectual lineage of this field extends considerably further than many realise. While modern research techniques have only recently been applied to these questions, philosophers and natural scientists have grappled with related issues for centuries. The transition from speculative philosophical inquiry to systematic empirical investigation represents one of the most significant developments in the intellectual history of this domain.

C The mechanisms underlying these phenomena operate at multiple levels of analysis, from the micro-level of individual decision-making to the macro-level of institutional and structural change. At the individual level, cognitive processes, emotional responses, and social influences shape the choices that people make. At the organisational level, formal rules, informal norms, and power dynamics govern collective action. At the societal level, historical legacies, cultural values, and political structures create the broad context within which lower-level processes unfold.

D The empirical record is further strengthened by a series of meta-analyses that have synthesised findings from hundreds of individual studies. These comprehensive reviews have consistently identified moderate to large effect sizes, with the most robust findings demonstrating remarkable consistency across different research designs, measurement approaches, and analytical strategies. This convergence of evidence from multiple independent lines of inquiry considerably strengthens confidence in the validity of the core theoretical propositions.

E Several unresolved debates continue to animate scholarly discussion in this field. Perhaps the most prominent concerns the relative importance of structural versus agentic factors in determining outcomes. While some researchers emphasise the constraining power of institutional and economic structures, others highlight the capacity of individuals and groups to exercise agency and effect change even within seemingly restrictive contexts. This debate has significant practical implications, as it shapes recommendations for policy interventions and institutional reforms.

F Looking ahead, the field stands at a critical juncture. Emerging technologies offer unprecedented opportunities for data collection and analysis, potentially enabling researchers to address questions that have long remained beyond the reach of empirical investigation. At the same time, the accelerating pace of social and technological change means that the phenomena under study are themselves evolving, requiring theoretical frameworks that can accommodate dynamism and discontinuity rather than assuming stability and equilibrium.

G The evidence reviewed above demonstrates both the substantial achievements of research in this field and the significant work that remains to be done. As new tools and theoretical perspectives become available, and as the phenomena themselves continue to evolve, the research community will need to maintain both rigour in its methods and openness to new ideas. The stakes are high, for the questions addressed by this research bear directly on some of the most important challenges facing humanity in the twenty-first century.

Question 1

Choose the correct letter, A, B, C or D.

1 What finding emerged from the landmark study discussed in the passage?

- A. excessive funding
- B. theoretical and methodological limitations
- C. public interest
- D. international collaboration

Question 2

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

2 The author believes that all research questions in this field have been answered.

Question 3

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

3 Dynamic _____ processes can amplify or dampen initial effects in complex systems.

Questions 4–6

Choose the correct letter, A, B, C or D.

4 Which of the following best describes the author's view on future developments?

- A. too much data collection
- B. the generalisability of findings across contexts
- C. insufficient government funding
- D. overly complex statistical methods

5 According to the text, what role do structural factors play?

- A. minimal impact
- B. creating underlying conditions for action
- C. determining individual choices directly
- D. irrelevant to outcomes

6 According to the text, what role do structural factors play?

- A. too much data collection
- B. the generalisability of findings across contexts
- C. insufficient government funding
- D. overly complex statistical methods

Question 7

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

7 The passage identifies power _____ as a key factor at the organisational level.

Questions 8–9

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

8 Meta-analyses have confirmed the reliability of the core findings in this field.

9 Contemporary understanding of this topic is more sophisticated than it was fifty years ago.

Question 10

Choose the correct letter, A, B, C or D.

10 What finding emerged from the landmark study discussed in the passage?

- A. inconsistent results across cultures
- B. a consistent pattern across diverse contexts
- C. no measurable effects
- D. effects limited to Western countries

Question 11

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

11 Early researchers had access to the same technologies as modern scholars.

Questions 12–13

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

12 The landmark study analysed data spanning a period of _____ years.

13 Case studies have documented the effects of _____ change on the system.